Taxless Finance and Its Consequences: New York and Pennsylvania, 1813–1860*

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Abstract

In the early 1840s multiple U.S. states defaulted on bonds issued to finance infrastructure projects, precipitating a wave of constitutional reforms that fundamentally reshaped state fiscal institutions. This paper examines New York and Pennsylvania as the "good" and "bad" case studies in that episode. Drawing on annual fiscal records for New York (1813–1860) and Pennsylvania (1820–1860), we trace how both states pursued "taxless finance"—borrowing against anticipated toll revenues without raising current taxes—and how their differing fiscal structures led to sharply different outcomes: New York narrowly avoided default while Pennsylvania defaulted in 1842. We then document the constitutional responses adopted by both states that eliminated the taxless-finance option and established fiscal constraints still influential in American public finance today.

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Contents

1 Introduction **4**

2 Analytical Framework: Government Budget Constraints **4**

2.1 New York: Three Budget Constraints . . . 5

2.2 Pennsylvania: One Budget Constraint . . . 5

3 New York State, 1813–1860: The "Good" Case **5**

3.1 Overview . . . 5

3.2 State Funds . . . 6

3.2.1 Origins: Hamilton's 1790 Settlement . . . 6

3.2.2 The Six Major Funds . . . 7

3.2.3 The Bank Safety Fund . . . 7

3.3 New York State Debts . . . 9

3.3.1 General Fund Debt and the Assumption of Railroad Obligations . . . 9

3.4 The Canal Account . . . 9

3.5 The General Account . . . 11

3.6 Consolidated Accounts . . . 15

3.7 Bond Prices and Fiscal Stress . . . 15

3.8 Unit of Account Issues . . . 15

4 Pennsylvania, 1820–1860: The "Bad" Case **17**

4.1 Overview . . . 17

4.2 Revenues and Expenditures . . . 18

4.3 Debt and Assets . . . 20

4.4 The Pennsylvania Canal and Railroad System . . . 20

4.5 The 1841–1844 Default Crisis . . . 23

4.5.1 Approach to Crisis . . . 23

4.5.2 1841: Relief Notes . . . 24

4.5.3 1842–1844: Interest Certificates and Asset Sales . . . 24

4.5.4 The Discrimination Problem . . . 24

4.6 Bond Prices . . . 25

4.7 Unit of Account Issues . . . 25

5 The Crisis in Comparative Perspective **25**

5.1 Other Defaulting States . . . 25

5.2 Patterns of Default . . . 26

6 Constitutional Responses **27**
6.1 New York Constitution of 1846 27
6.2 Pennsylvania Constitutional Amendments of 1857 27
6.3 Common Elements Across the Reforms 28

7 The "Taxless Finance" Problem **29**

8 Long-Run Impact **29**

9 Conclusion **30**

10 Data Sources **30**

1 Introduction

In the early 1840s, the United States experienced a sovereign debt crisis at the state level. Nine states and one territory defaulted on bond obligations following the Panic of 1837, marking one of the most significant fiscal crises in American history. This crisis led to fundamental constitutional reforms in multiple states, with New York and Pennsylvania among those adopting strict limitations on state borrowing and prohibitions on the pledging of state credit to private enterprises.

The defaults were concentrated among states that had borrowed heavily to finance "internal improvements"—primarily canals and railroads—using what economic historians term *taxless finance*. Rather than raising property taxes to fund infrastructure, states borrowed against anticipated toll revenues and hoped that increased land values would eventually generate sufficient tax receipts to service the debt (Wallis 2005). When the economic depression following the Panic of 1837 reduced both toll revenues and land values, states that had relied on this strategy found themselves unable to service their debts.

New York and Pennsylvania make ideal paired case studies. Both states invested heavily in canal systems in the 1820s and 1830s; both accumulated large debts; both faced severe fiscal pressure in the early 1840s. Yet their outcomes differed sharply. New York—whose Erie Canal was an extraordinary commercial success—narrowly avoided default, while Pennsylvania defaulted in 1842 and remained in default until 1845. The contrast illuminates both the role of project quality and the role of pre-existing fiscal institutions.

The remainder of this paper is organized as follows. Section 2 presents the government budget constraints that organize our fiscal analysis. Sections 3 and 4 provide detailed fiscal portraits of New York and Pennsylvania, respectively. Section 5 places the two states in comparative perspective. Section 6 documents the constitutional responses. Section 7 analyzes the taxless-finance problem. Section 8 discusses long-run impact, and Section 9 concludes.

2 Analytical Framework: Government Budget Constraints

The organizing principle for our fiscal analysis is the government budget constraint with private assets ("state funds"):

$$\underbrace{G_t}_{\substack{\text{government}\\\text{spending}}} + \underbrace{r_t^B B_{t-1}}_{\substack{\text{interest on}\\\text{state debt}}} + \underbrace{A_t - A_{t-1}}_{\substack{\text{change in}\\\text{state funds}}} = \underbrace{T_t}_{\text{taxes}} + \underbrace{r_t^A A_{t-1}}_{\substack{\text{interest on}\\\text{state funds}}} + \underbrace{B_t - B_{t-1}}_{\substack{\text{change in}\\\text{state debt}}} \tag{1}$$

This consolidated constraint holds for both states, though their accounting structures differ: New York maintained separate "Canal" and "General" accounts, while Pennsylvania recorded public-works spending and revenues within a single general account.

2.1 New York: Three Budget Constraints

New York's separate Canal Commission accounting requires us to decompose the consolidated constraint (1). The *Canal Account* satisfies:

$$\begin{aligned}\underbrace{F_t^c - F_{t-1}^c}_{\substack{\text{change in}\\ \text{Canal Fund}}} &= \underbrace{B_t^c - B_{t-1}^c}_{\text{new Canal Debt}} + \underbrace{TOLL_t^c}_{\text{tolls}} + \underbrace{T_t^c}_{\text{taxes}} + \underbrace{OR_t^c}_{\text{other revenue}} + \underbrace{TR_t^{gc}}_{\substack{\text{transfers from}\\ \text{General Account}}} \\ &+ \underbrace{r_t^A F_{t-1}^c}_{\substack{\text{interest on}\\ \text{Canal Fund}}} - \underbrace{EXP_t^c}_{\text{Canal expenses}} - \underbrace{r_t^B B_{t-1}^c}_{\substack{\text{interest on}\\ \text{Canal debt}}} - \underbrace{TR_t^{cg}}_{\substack{\text{transfers to}\\ \text{General Account}}} \end{aligned} \tag{2}$$

The *General Account* satisfies:

$$\begin{aligned}\underbrace{F_t^g - F_{t-1}^g}_{\substack{\text{change in}\\ \text{General Fund}}} &= \underbrace{B_t^g - B_{t-1}^g}_{\substack{\text{change in}\\ \text{General Debt}}} + \underbrace{T_t^g}_{\text{taxes}} - \underbrace{TR_t^{gc}}_{\substack{\text{transfer to}\\ \text{Canal Account}}} + \underbrace{r_t^A F_{t-1}^g}_{\substack{\text{interest on}\\ \text{General Fund}}} \\ &- \underbrace{S_t^g}_{\substack{\text{General Account}\\ \text{spending}}} - \underbrace{r_t^B B_{t-1}^g}_{\substack{\text{interest on}\\ \text{General Fund debt}}} + \underbrace{TR_t^{cg}}_{\substack{\text{transfers from}\\ \text{Canal Account}}} \end{aligned} \tag{3}$$

Adding (2) and (3) and defining $A_t = F_t^c + F_t^g$, $B_t = B_t^c + B_t^g$, $T_t = TOLL_t^c + T_t^c + T_t^g + OR_t^c$, $G_t = S_t^g + EXP_t^c$ recovers the consolidated constraint (1).

2.2 Pennsylvania: One Budget Constraint

Unlike New York, Pennsylvania included public-works spending and revenues directly in its general account, so the consolidated constraint applies directly with A_t capturing the Commonwealth's holdings of bank stocks, turnpike stocks, and canal and navigation stocks.

3 New York State, 1813–1860: The "Good" Case

3.1 Overview

New York presents the more favorable outcome in the state debt crisis. Several features distinguish it from states that defaulted:

- In the 1790s and early 1800s, New York legislators aspired to finance state government from permanent funds derived from land sales and special duties—a government "without taxation." Later, with the opening of the Erie Canal, the state became largely a venture-capital operation.
- New York paid off its pre-1814 debt by the mid-1820s but resumed borrowing to pay for ordinary expenditures in the 1830s.

- New York borrowed heavily to finance the construction and maintenance of the Canal System, with canals representing the dominant share of state debt throughout this period.
- In the late 1830s and early 1840s, New York faced two major fiscal shocks: (i) three railroad companies whose debt was guaranteed by the State went bankrupt, and (ii) a state-run bank insurance system collapsed when 11 banks failed.
- Despite these shocks, New York did not default on its debt during the crisis of the 1840s. Its bonds traded at a discount, and the Comptroller complained publicly—but coupon and principal payments were maintained.
- New York adopted a new constitution in 1846 that fundamentally changed the rules governing state borrowing.
- There are important connections to Hamilton's 1790 federal assumption of state war debts, which formed the foundation of New York's General Fund. There are also recurring unit-of-account issues during periods of paper currency.

3.2 State Funds

3.2.1 Origins: Hamilton's 1790 Settlement

In the 1780s, New York held substantial Revolutionary War debts. It also held the following Continental Securities:

State Holdings of Continental Securities	Amount (£ s d)
Loan Office Certificates	360,683 9 3
Certificates, Wm. Barber	458,213 4 0
Certificates, Pierce, Burrel, Deming	389,498 9 9
Indent of Interest	2,502 14 8
Total	1,210,897 16 8

In 1791, Alexander Hamilton exchanged these securities for:

United States 6% Stock	$898,847.03
United States deferred 6% stock	$520,688.66
United States 3% Stock	$680,681.70
Total	$2,100,197.39

(Source: Sowers (1914, pp. 255–6).) The accounts between the State and the Secretary of the Treasury were finally settled in 1800.

These three stocks formed the basis of what became known as the *General Fund*. The Comptroller was made custodian and was empowered to invest it; over time, the fund was augmented from sales of public lands and other sources.

3.2.2 The Six Major Funds

Between 1813 and 1860, New York maintained six major funds and eight minor sinking funds:

1. The *General Fund* was created from the 1790 federal settlement and augmented with asset sales; it helped fund ordinary government but its principal was gradually drawn down and ultimately replaced by debt.

2. The *Canal Fund* was financed by the revenues of the Erie Canal and other canals and used to pay canal expenses and Canal Fund debts.

3. The *Common School Fund* was a permanent endowment fund for public education, established in 1805 from the proceeds of the first 500,000 acres of vacant land managed by the Surveyor-General. No direct state tax for school purposes was levied until 1853; from 1805 to 1853, the interest on the fund—supplemented by the U.S. Deposit Fund—supported the state's system of free public education. The fund grew from $0.82 million in 1813 to $2.6 million in 1860.

4. The *Literature Fund* (est. 1827) supported secondary education and the creation of libraries in *academies*. Unlike the Common School Fund, which supported free local schools, the Literature Fund served privately governed academies charging tuition. It was eventually consolidated with the U.S. Deposit Fund.

5. The *Bank Safety Fund* (described below) was a bank insurance fund that went insolvent after 11 banks failed between 1840 and 1842.

6. The *U.S. Deposit Fund* was created after the federal government distributed its fiscal surplus to the states under the Deposit Act of 1836. The money was lent by the state secured by real-estate mortgages; interest supported the Common School Fund.

The eight minor sinking funds were created following the state's assumption of debts from three failed railroads (see Section 3.3).

In Figure 1, we plot the balances for these funds.

3.2.3 The Bank Safety Fund

In 1829, New York established a bank insurance system, the *Bank Safety Fund*. Banks were required to deposit securities in the Comptroller's office as guaranty for their circulating notes, with annual assessments of $\frac{1}{2}$ of 1% of paid-in capital until each bank had contributed 3%. The fund would reimburse creditors of any failing bank.

Albert Gallatin observed presciently:

> ... with the laudable intent to protect the community against partial failures, a 'safety fund' has been established by law ... which is applicable to the payments of the notes of any that may fail. This must have the tendency to encourage excessive issues of paper,

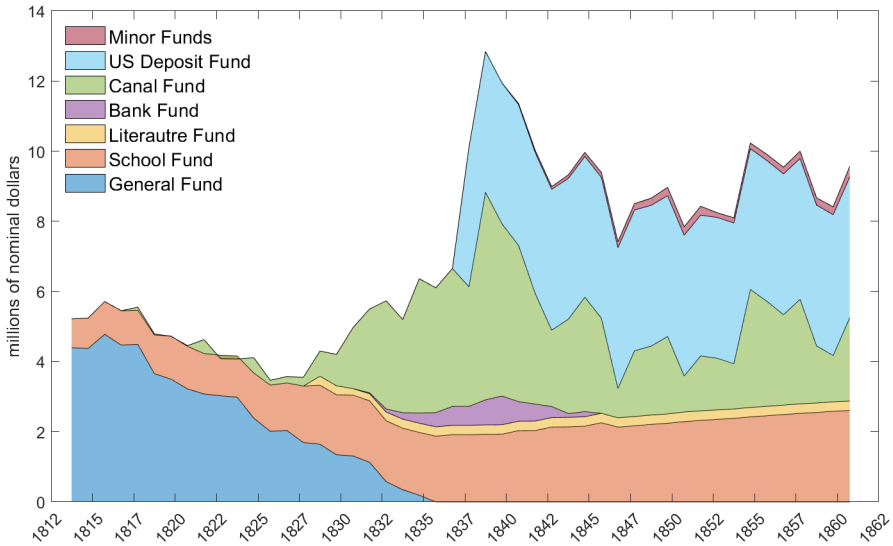

Figure 1: New York State Fund Balances

which could not be sustained if resting only on the credit of the bank by which they are made. (Bodenhorn 1996)

Between February 1840 and October 1842, eleven banks failed (Table 1). The total cost of insuring their creditors was $2.565 million; the fund became insolvent in 1843.

Failed Bank	Date of Failure	Notes Redeemed	Other Debts Paid	Receipts from Asset Sales	Balance Paid by Fund
City Bank of Buffalo	2/1840	$317,107		$99,996	$217,111
Wayne County Bank	12/1840	113,131	$16,078		129,209
Commercial Bank of New York City	9/1841	139,837	146,129	$7,188	278,778
Bank of Buffalo	11/1841	435,540	149,241		584,781
Commercial Bank of Buffalo	11/1841	186,861	424,515	5,000	606,376
Commercial Bank of Oswego	12/1841	163,162	78,352	2,392	239,122
Watervliet Bank	3/1842	134,107	77,484	13,258	198,333
Clinton County Bank	4/1842	71,896	156,257		228,153
Bank of Lyons	9/1842	52,898	40,053	3,761	88,990
Lafayette Bank, NYC	2/1842	38			38
Oswego Bank	10/1842	725		6,482	-
Total		$1,615,302	$1,088,109	$138,077	$2,565,334

Table 1: Payments Made by the Bank Safety Fund to Reimburse Creditors of the Failed Banks

Source: Chaddock (1910, p. 332) and Bodenhorn (1996, p. 25).

3.3 New York State Debts

Figure 2 plots the quantity of New York State debt outstanding. There are three types: General Fund debt, Contingent debt, and Canal debt. Canal debt constitutes the lion's share. About 42% of the total was held abroad. Table 2 reproduces the ownership distribution from the 1842 Annual Report of the Comptroller.

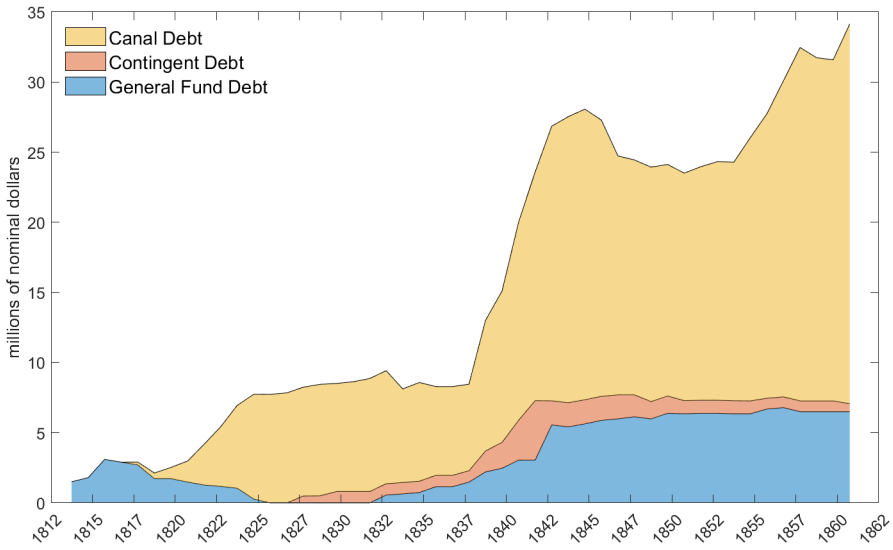

Figure 2: New York State Debt

3.3.1 General Fund Debt and the Assumption of Railroad Obligations

General Fund debt appeared in 1814 alongside a positive General Fund balance; it was paid off by 1825. In 1832 the State resumed borrowing to cover ordinary expenditures.

Starting in 1827, the State issued *contingent debt*: bonds issued on behalf of canal and railroad companies for which the State pledged its faith and credit. On July 1, 1841, the Ithaca and Owego and Catskill and Canajoharie railroads ceased coupon payments; the New York and Erie Railroad subsequently defaulted as well. As a result, $3,315,700 in railroad debt became a direct State obligation.

3.4 The Canal Account

New York's Erie Canal, completed in 1825, was the dominant source of state revenue and the primary driver of state debt. Figures 3–5 plot the Canal Account's revenues, expenditures, and their comparison.

- From 1817 to 1820, taxes on auctions, salt, and steamboat revenue constituted the sole Canal Account revenue. Between 1817 and 1836 these taxes were diverted from the General Account

Bond	Held in New York	Held in Other States	Held by Foreigners	Total
Erie and Champlain Canals	$150,530	$18,495	$1,715,412	$1,884,437
Enlargement of Erie Canal	3,294,049	333,229	3,407,341	7,034,619
Cayuga, Seneca, and Oswego Canals	376,079	26,211	256,014	658,304
Crooked Lake Canal	114,000	—	5,500	120,000
Chemung Canal	156,007	5,700	314,707	476,974
Chenango Canal	629,232	223,392	1,549,911	2,402,536
Black River Canal	357,552	8,167	739,955	1,105,673
Genesee Valley Canal	606,064	81,905	1,878,411	2,566,380
Oneida River Improvement	43,800	1,200	5,000	50,000
Oneida Lake Canal	50,000	—	—	50,000
7% stock of 1848	1,445,736	106,500	32,500	1,584,736
7% stock of 1849	1,608,000	13,600	31,000	1,652,600
Delaware & Hudson Canal Co.	409,317	17,359	373,324	800,000
New York & Erie Railroad Co.	2,587,000	251,000	162,000	3,000,000
Tonawanda Railroad	86,000	14,000	—	100,000
Schenectady Railroad	100,000	—	—	100,000
Long Island Railroad	80,000	3,000	17,000	100,000
Catskill and Canajoharie Railroad	86,000	—	114,000	200,000
Ithaca and Owego Railroad	82,000	12,000	221,700	315,700
Auburn and Rochester Railroad	188,000	11,000	1,000	200,000
Hudson and Berkshire Railroad	141,000	—	9,000	150,000
Tioga Coal, Iron and Mining Co.	70,000	—	—	70,000
Astor Stock	561,500	—	—	561,500
5% Bank Fund Stock	350,257	—	—	350,257
7% stock and bonds	465,358	—	—	465,358
Total	$14,038,540	$1,240,962	$10,833,776	$25,999,074

Table 2: Ownership Distribution of New York State Debts, 1841–42

Source: 1842 *Annual Report of the Comptroller*, pp. 9–10.

to the Canal Account.

- Although the Erie Canal officially opened on October 26, 1825, toll collection began in 1821. By the late 1820s tolls became the Canal's largest revenue source; revenues declined in the late 1830s and 1850s.
- Prior to 1840, the Canal Account received transfers from the General Account. After 1840, the flow reversed: the Canal Account transferred surpluses back to the General Account (Figure 6).

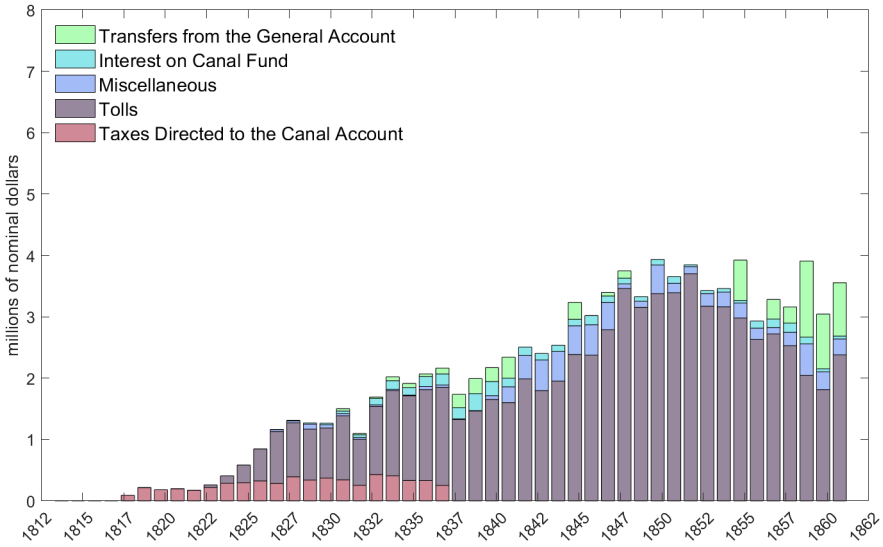

Figure 3: New York Canal Account Revenues

3.5 The General Account

- A state property tax was in effect between 1816 and 1826, accounting for a large share of General Account revenue. It was reinstated in 1842 in response to the fiscal crisis.
- During the late 1820s and early 1830s, revenue from the state accounts (primarily returns on the General Fund investment portfolio) made up the lion's share of General Account revenue. In 1832 the State resumed borrowing for ordinary expenditures, signaling that investment income was no longer sufficient.
- The 1836 distribution of the federal surplus under the Deposit Act provided a one-time revenue boost.
- The assumption of contingent railroad debt in 1841–42 and the Bank Safety Fund payments to failed-bank creditors both raised General Account outlays sharply.

Figures 7–9 plot General Account revenues, expenditures, and their comparison.

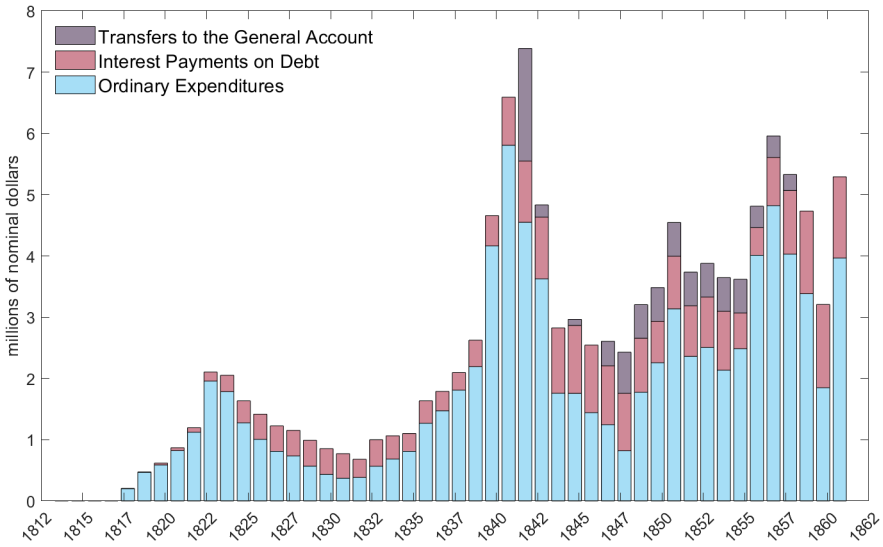

Figure 4: New York Canal Account Expenditures

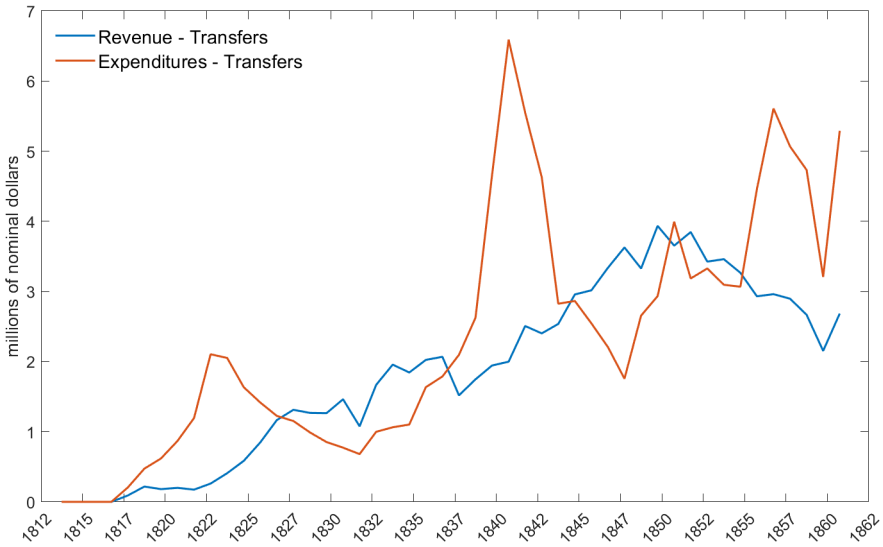

Figure 5: New York Canal Account Revenue and Expenditures (net of inter-account transfers)

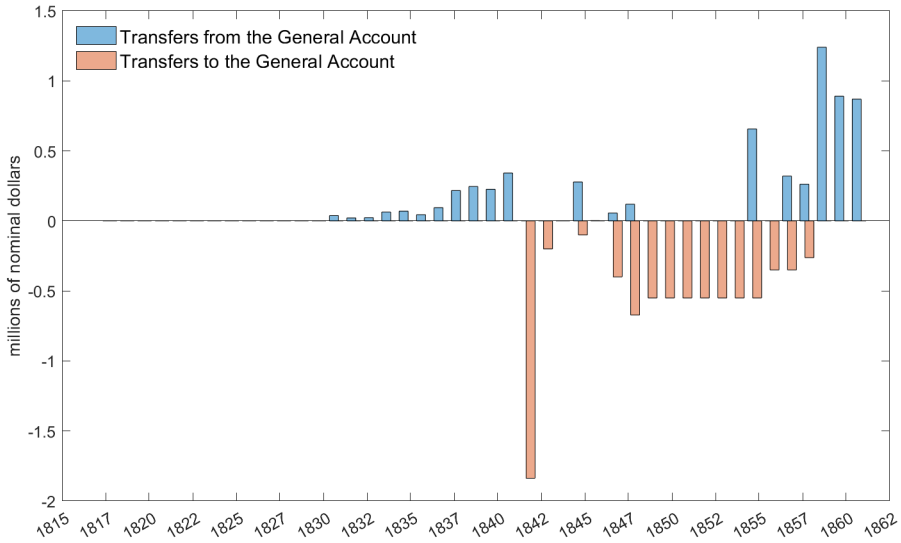

Figure 6: Transfers Between the Canal and General Accounts

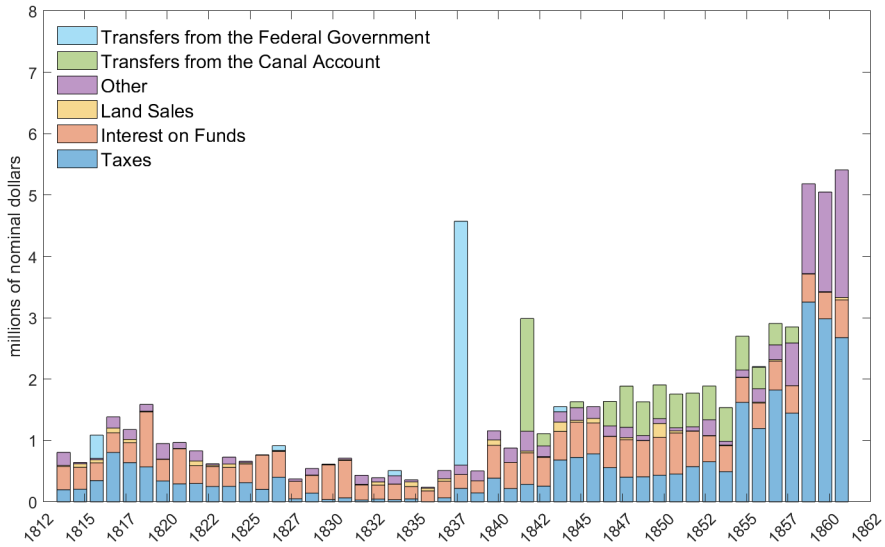

Figure 7: General Account Revenues

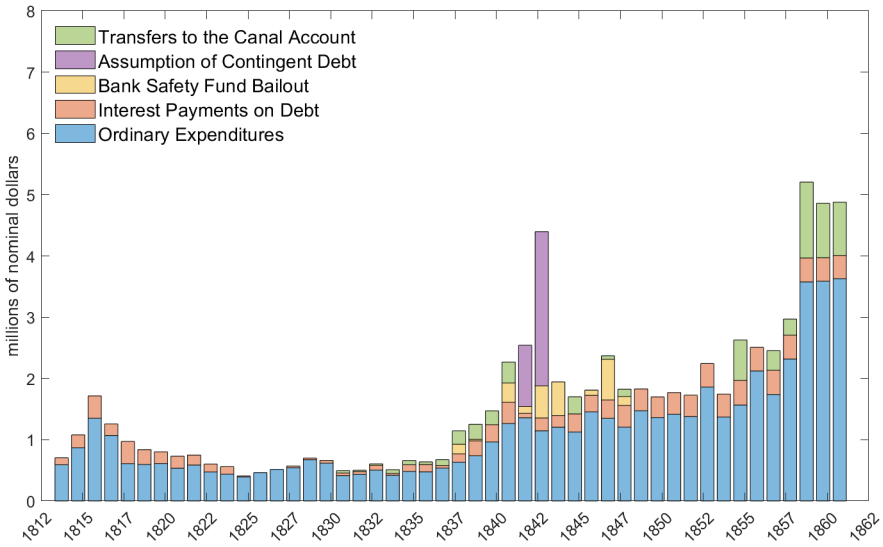

Figure 8: General Account Expenditures

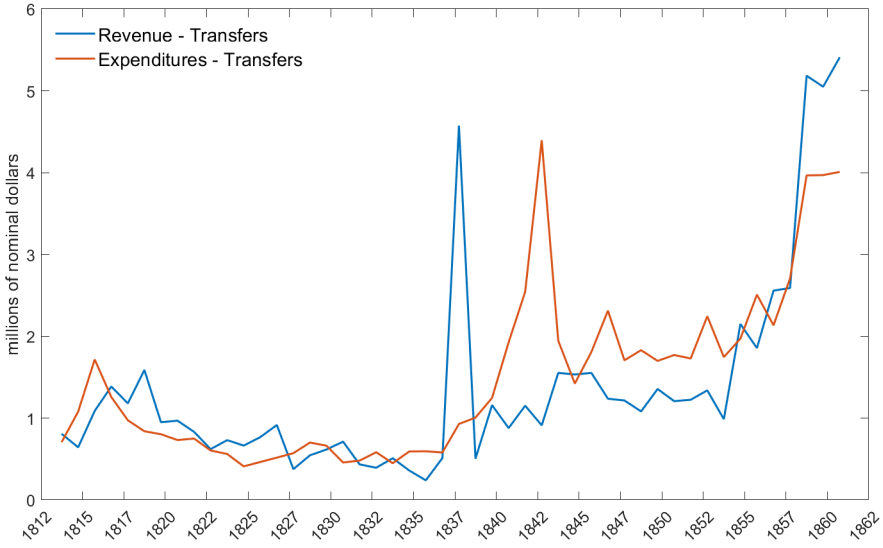

Figure 9: General Account Revenue and Expenditures

3.6 Consolidated Accounts

Figures 10–12 merge the Canal and General Accounts. The Canal Account's revenues and expenditures dwarf those of the General Account: New York State was primarily a canal operator with a secondary role in ordinary government. With the important exception of the late 1830s and early 1840s, the consolidated budget was broadly balanced.

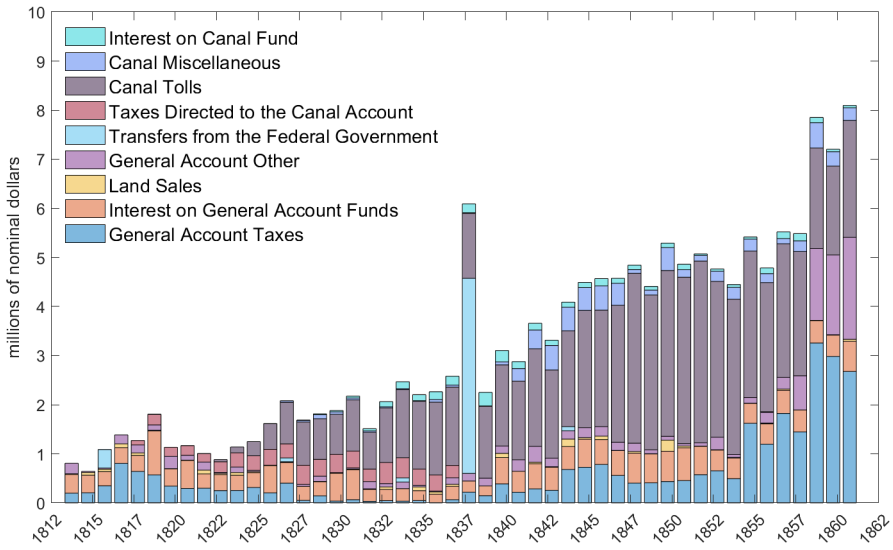

Figure 10: New York Consolidated Revenues

3.7 Bond Prices and Fiscal Stress

Figure 13 plots the prices of three Canal bonds and three General Fund bonds.

Despite never missing a coupon or principal payment, New York State bonds traded at 80 cents on the dollar in the early 1840s. Comptrollers complained publicly about the discount. The late 1830s and early 1840s were a period of acute fiscal stress: the Legislature suspended all public works except those whose suspension would cause a significant loss, and it pledged the state's revenues for the payment of its obligations.

3.8 Unit of Account Issues

New York's fiscal record repeatedly intersected with monetary standard questions:

- When specie payments were suspended during the financial panic of 1837, Comptroller Azariah C. Flagg used State resources to make up the difference between depreciated paper currency and coin, thus honoring the State's obligations at par.
- In 1857, Comptroller William F. Allen urged that the State's bonded debt be serviced in coin rather than paper, arguing that the public faith of New York should not fluctuate with the

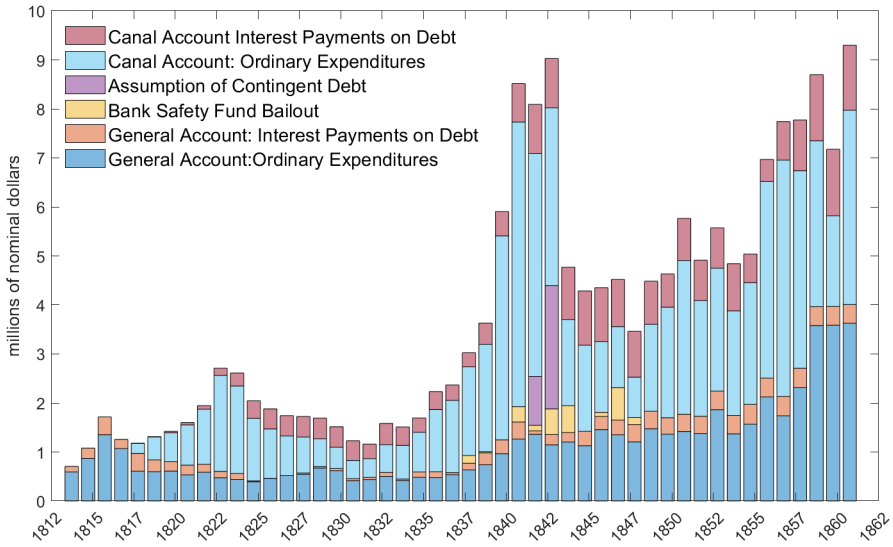

Figure 11: New York Consolidated Expenditures

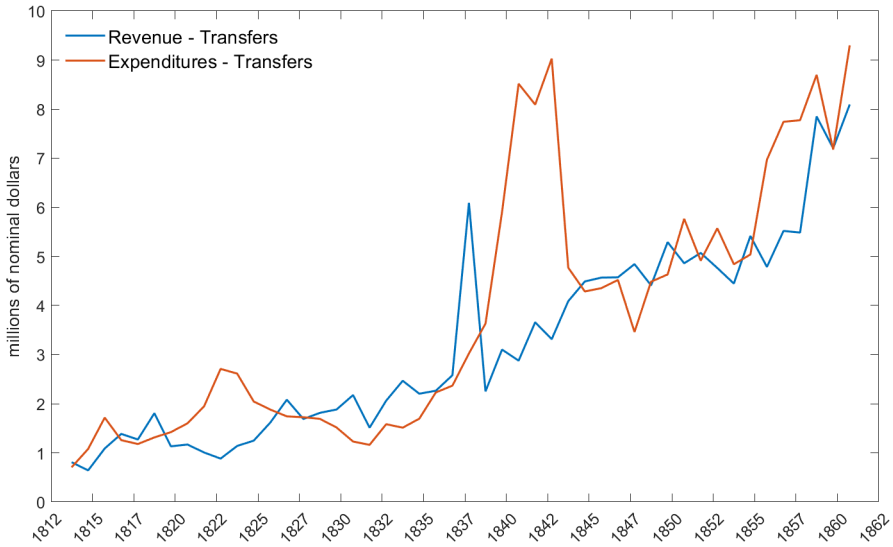

Figure 12: New York Consolidated Revenue and Expenditures

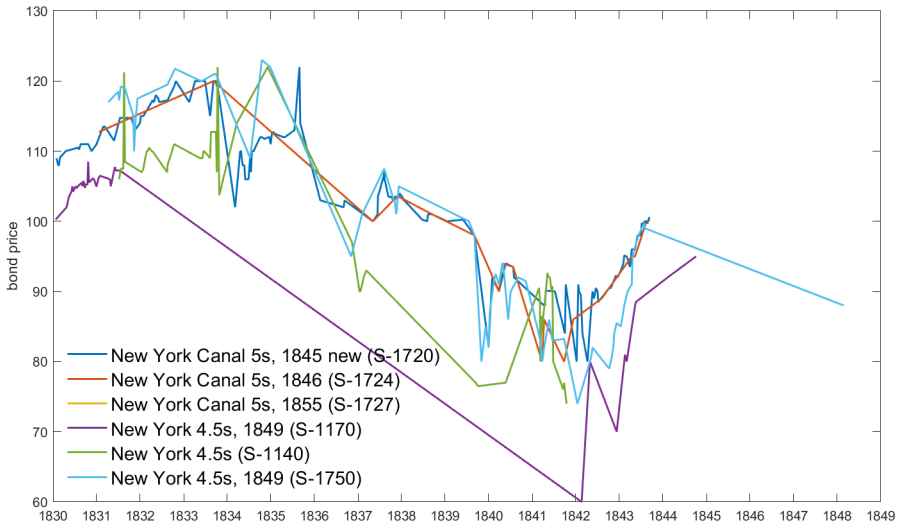

Figure 13: Prices of Various New York State Bonds

Source: Sylla et al. (2006).

market value of its currency.

- During the Civil War, when specie commanded a substantial premium over paper, Comptroller Lucius Robinson recommended (in 1863 and 1864) that both principal and interest on the State's bonds be paid in specie. The Legislature declined.

- In 1870, the State entered the open market to purchase coin for interest payments, a practice continued until the federal government's general resumption of specie payments in 1879. By adhering voluntarily to a metallic standard during the years of national paper currency, New York strengthened its credit-market standing and set a precedent for conservative financial management that endured well into the late nineteenth century.

4 Pennsylvania, 1820–1860: The "Bad" Case

4.1 Overview

Pennsylvania presents the unfavorable outcome. Inspired by the early success of New York's Erie Canal, Pennsylvania embarked on a more ambitious and ultimately less successful program of state-sponsored internal improvements. The Commonwealth borrowed heavily, maintained an aversion to direct taxation, and defaulted in August 1842. The contrast with New York captures in miniature the broader cross-state patterns identified by Wallis (2005).

Unlike New York, Pennsylvania included public-works spending and revenues directly in its general account, a feature that both simplified the accounting and obscured the true fiscal position.

4.2 Revenues and Expenditures

Figures 14–17 plot Pennsylvania's revenues and expenditures.

- Spending on internal improvements and interest payments constituted by far the largest expenditure categories.
- In 1842, 1843, and 1844, Pennsylvania failed to make promised interest and other payments. Bondholders were issued "interest certificates," while unpaid claims from government contractors were recorded as debts to "domestic creditors."
- The 1836–37 spike in revenue reflects the federal distribution of surplus funds under the Deposit Act of 1836, together with unusually large fees from bank-charter enrollments.
- After the crisis, the Commonwealth ran a primary surplus and achieved a broadly balanced gross budget. Tax revenue increased substantially starting in 1845 as a large number of new taxes were introduced.
- In 1840, Pennsylvania's debt stood at $33.3 million and the State collected $1.9 million in total revenue. That same year, New York's consolidated debt was $17.6 million and it collected $2.9 million in total revenue—a substantially better debt-to-revenue ratio.

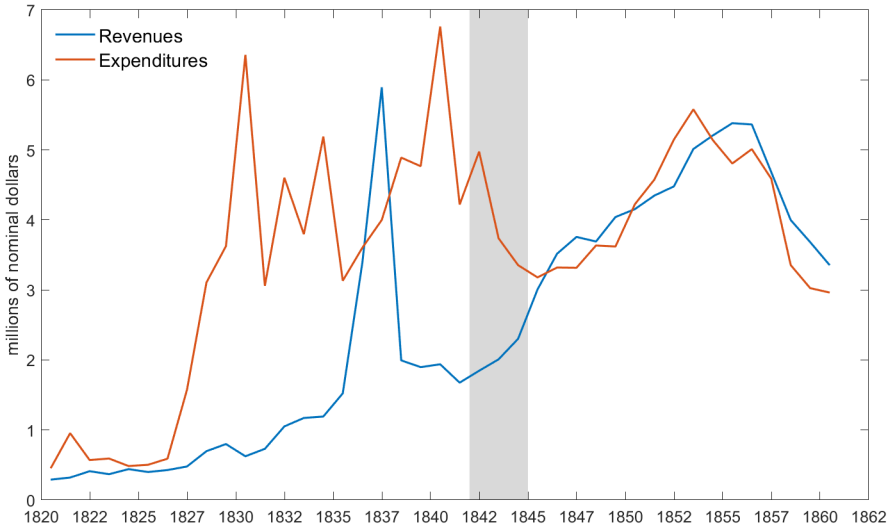

Figure 14: Pennsylvania State Revenue and Expenditures

The period of default (1842–1844) is shaded in grey.

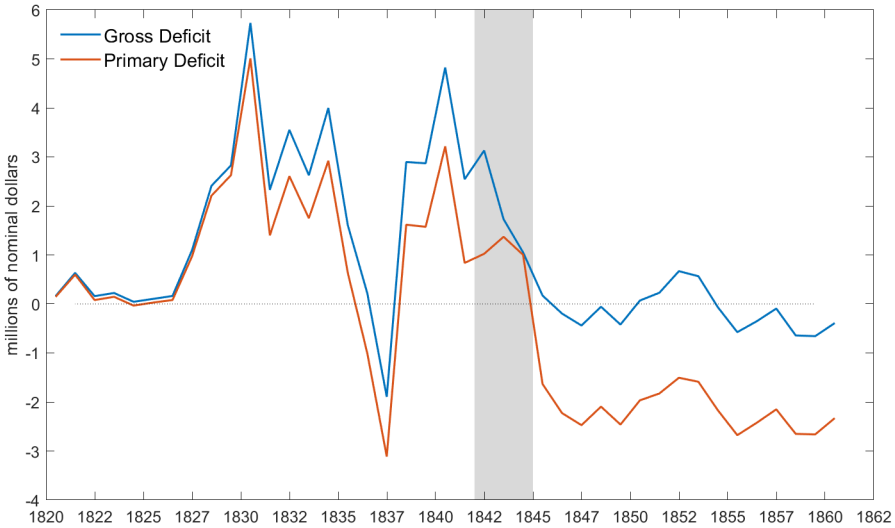

Figure 15: Pennsylvania Gross and Primary Deficits

The period of default (1842–1844) is shaded in grey.

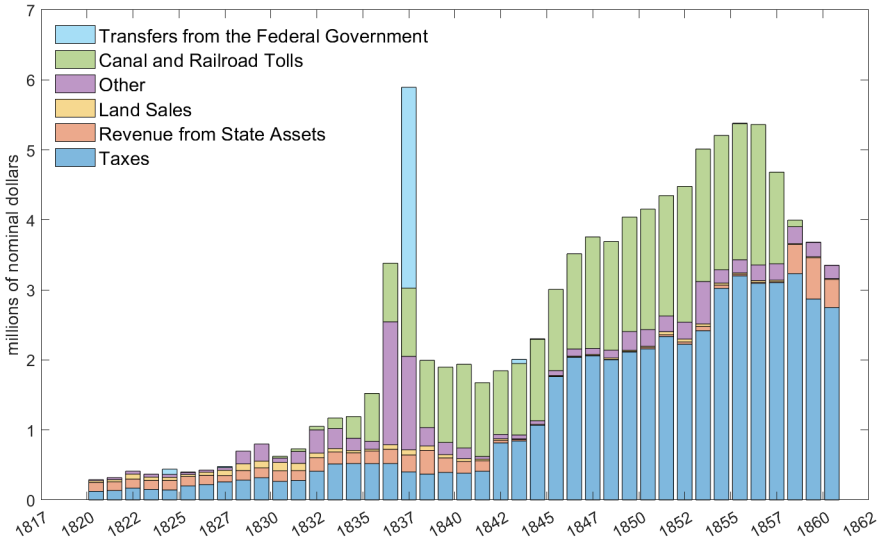

Figure 16: Pennsylvania Revenue Decomposed

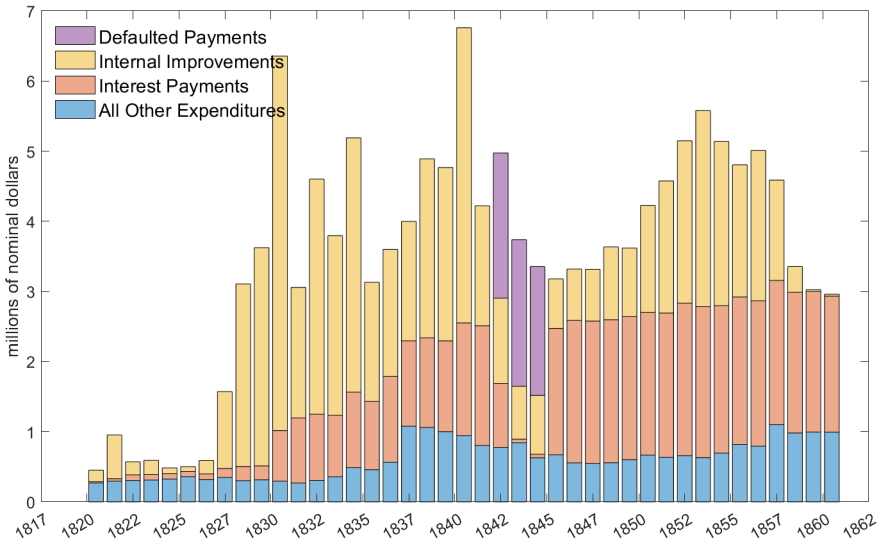

Figure 17: Pennsylvania Expenditure Decomposed

4.3 Debt and Assets

- Between 1825 and 1840, nearly all outstanding debt had been issued to fund internal improvements. Starting in the 1840s, nearly all new debt was issued to roll over past debt or cover coupon payments; these refinancing loans allowed other revenue to continue financing internal-improvement spending.

- Pennsylvania owned bank stocks, turnpike and bridge stocks, and canal and navigation stocks (Figure 19). In 1843, the State sold $4,169,440 face value of these stocks, receiving only $1,395,412— approximately one-third of face value.

- Starting in 1835, the State's canals, railroads, and bridges were recorded at cost as State Assets, but are not included in Figure 19.

4.4 The Pennsylvania Canal and Railroad System

In February 1826, the Pennsylvania Legislature authorized the Main Line of Public Works—a system of canals and railroads connecting Philadelphia to Pittsburgh—in direct response to the opening of New York's Erie Canal. Between 1830 and 1857 the State operated nine canal divisions and two railroads:

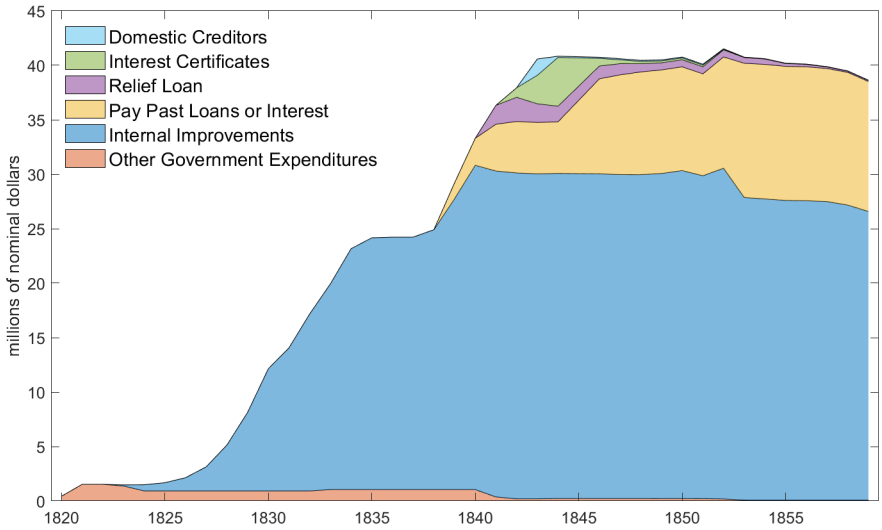

Figure 18: Pennsylvania Debt Decomposed by Purpose

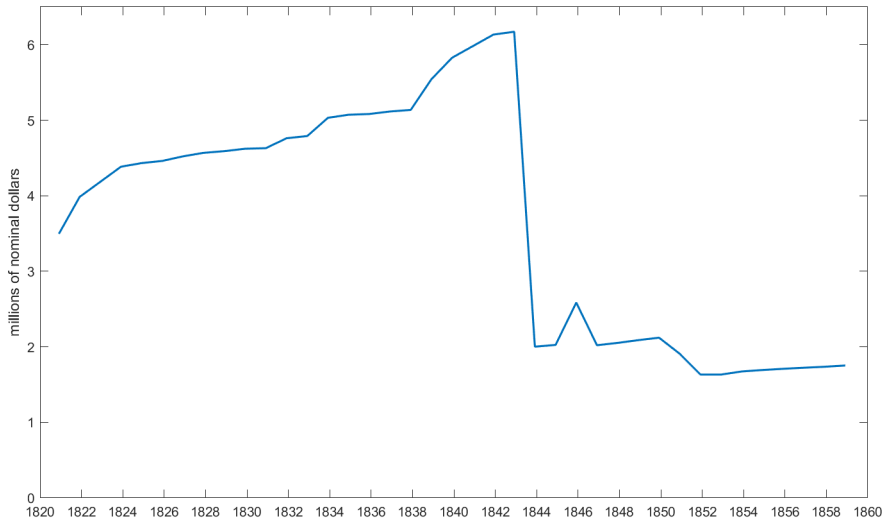

Figure 19: Pennsylvania State Assets, Not Including Public Works

	Length (miles)	Cost of Construction
Railways		
Columbia and Philadelphia	82	$5,277,278
Allegheny Portage	36	2,708,672
Railway Total	118	$7,985,950
Pennsylvania Canal		
Eastern Division	45	$1,737,285
Juniata Division	128	3,575,966
Western Division	103	3,173,432
Delaware Division	60	1,454,937
Susquehanna Division	41	897,161
North Branch Division	73	1,598,379
West Branch Division	76	1,832,583
French Creek Division	49	817,780
Beaver Division	30	519,365
Canal Total	605	$15,606,888
Unfinished Improvements		$8,695,044
Other Costs		254,384
Grand Total		$32,542,268

Source: Pennsylvania Office of the Auditor General and Treasury (1854).

The average construction cost of the Pennsylvania Canal was $25,797 per mile, compared with $19,678 per mile for the Erie Canal. The comparison highlights that the Pennsylvania network was both more expensive per mile and—as the revenue figures show—far less commercially successful.

Table 3 summarizes the fiscal outcome over the life of the public works. The losses were staggering: between 1820 and 1853, the State spent between $84 million (our estimate) and $87 million (the official estimate) to build, finance, and operate these works, while collecting only $25 million in toll revenue.

	Official Numbers	Our Numbers
Running Expenses, 1830–1853	$19,447,654	$19,477,654
Interest Payments on Canal & Railroad Debt, 1827–1853	35,157,796	35,231,591
Construction Costs, 1820–1853	32,542,268	28,954,372
Total Costs	$87,147,717	$83,663,617
Revenue from Tolls	$24,995,632	$24,995,632

Table 3: Aggregated Costs and Revenue for the Pennsylvania Canals and Railroads

Source: Pennsylvania Office of the Auditor General and Treasury (1854).

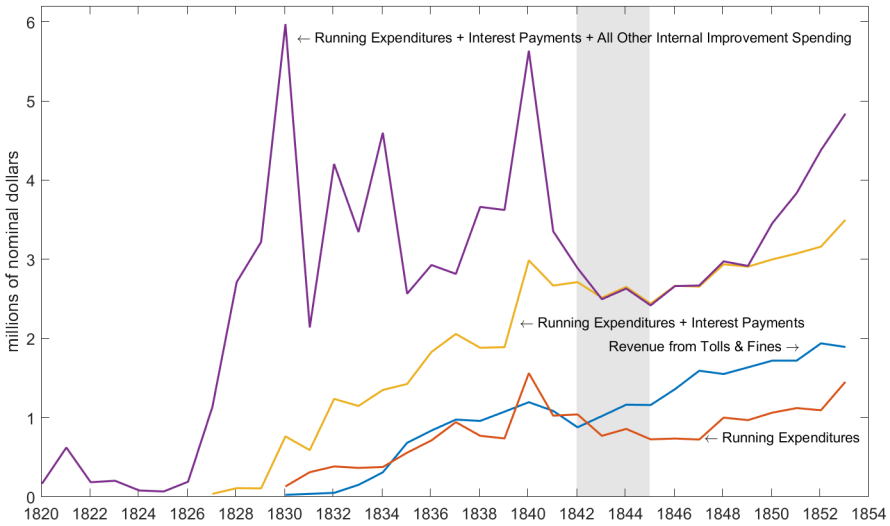

Figure 20: Pennsylvania Canal and Railroad Revenue and Expenditures

The period of default (1842–1844) is shaded in grey.

After 1842, toll revenues consistently exceeded running expenses, but fell nowhere near covering interest payments—much less recouping construction costs. In 1844 the State offered the works for sale at $20 million; there were no takers. In 1854 another sale attempt likewise failed. Eventually:

- On June 25, 1857, the Columbia and Philadelphia Railway, the Allegheny Portage Railway, and several canal divisions were sold to the Pennsylvania Railroad Company for $7.5 million.
- In 1858, the remaining canals and railroads were sold to the Sunbury and Erie Railroad Company for $3.5 million.

4.5 The 1841–1844 Default Crisis

4.5.1 Approach to Crisis

In 1838, 1839, and 1840, Pennsylvania ran budget deficits of $2.9, $2.9, and $4.8 million, respectively. In 1841, the deficit was $2.5 million: expenditures totaled $4.2 million (including $675,000 in loan repayments, $1,707,000 in interest, $1,710,000 in internal improvements, and $803,000 in ordinary expenditures) against only $1.7 million in revenue. The State was shut out of the bond market.

Figure 21 plots the debt service profile as of December 1840. Pennsylvania did not face large principal maturities in the 1840s—it was the promised coupon payments that proved fatal.

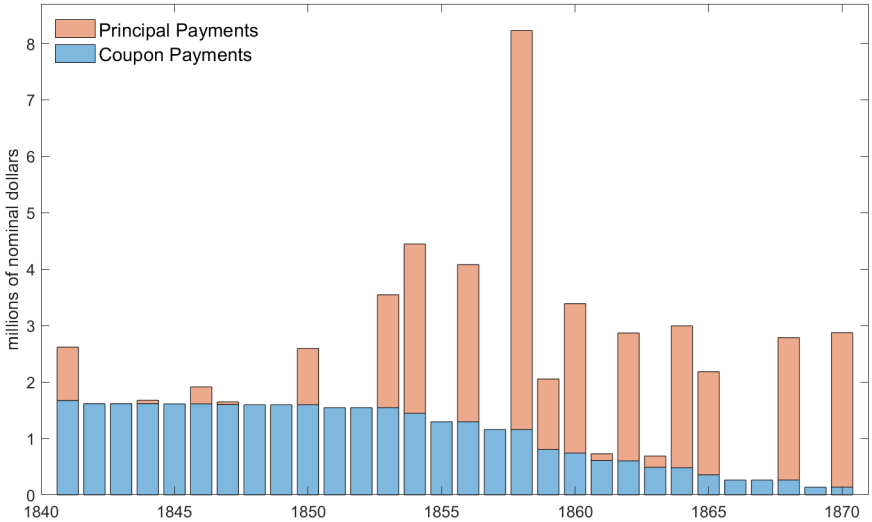

Figure 21: Debt Service Profile for December 1840

4.5.2 1841: Relief Notes

To meet its 1841 obligations without outright default, Pennsylvania passed the Relief Note Act of May 4, 1841, authorizing a $2.22 million"relief loan." Banks that subscribed were permitted to issue small-denomination paper currency ("relief notes" in $1, $2, and $5 denominations, normally illegal) up to their subscription amount, and were granted relief from certain state taxes and penalties for suspending specie payments. The notes were made receivable for all debts due to the Commonwealth, effectively making them a state-backed currency.

4.5.3 1842–1844: Interest Certificates and Asset Sales

In February 1842 Pennsylvania met its interest payment, but by August 1 the treasury was exhausted. English (1996) dates the formal default as August 1842 and the resumption as February 1845.

The State Treasurer issued to bondholders *certificates of stock* equal to interest then due, bearing 5% interest and payable on August 1, 1843. Government contractors and public-works employees unpaid during the crisis were classified as *domestic creditors*. County commissioners were instructed to levy an additional tax of one mill on the dollar on all real and personal property.

In April 1843, the State sold all of its holdings in bank stocks and other incorporated companies. Face value was $4,169,440; the State received $1,395,412—one-third of face value.

4.5.4 The Discrimination Problem

The default forced the Commonwealth to choose among:

1. Failing to pay domestic creditors (contractors, employees),

2. Failing to pay foreign bondholders, or

3. Raising taxes on Pennsylvania citizens.

This was an echo of the discrimination problem Hamilton faced in 1790 when designing the federal assumption of state debts. Failing domestic creditors or raising taxes risked immediate electoral backlash and social unrest; failing foreign bondholders carried reputational costs that were diffuse and delayed. In 1842, of the $34,454,356 in Pennsylvania debt outstanding, approximately 69% was held by foreigners (overwhelmingly English):

Pennsylvania citizens	$9,635,613
Other U.S. citizens	1,080,538
Foreigners	23,738,206
Total	$34,454,357

Source: Pennsylvania Office of the Auditor General (1842).

The five largest foreign-creditor countries were England ($20,026,458), Holland ($1,822,266), France ($570,000), the West Indies ($563,161), and Switzerland ($239,677). No formal restructuring occurred; the exact mechanism by which the discrimination problem was resolved remains to be fully established.

4.6 Bond Prices

Figure 22 plots Pennsylvania bond prices. The contrast with Figure 13 for New York is instructive: Pennsylvania bonds fell further and recovered more slowly, reflecting the deeper fiscal distress and the actual interruption of payments.

4.7 Unit of Account Issues

In May 1837, Pennsylvania banks suspended specie payments in response to the financial panic. In July 1839, the Legislature resolved that the State would pay interest in "lawful money" (specie), and the Treasury records show evidence of making up differences between paper and specie values. Pennsylvania's unit-of-account management stands in contrast to New York's more consistent adherence to a metallic standard.

5 The Crisis in Comparative Perspective

5.1 Other Defaulting States

Beyond Pennsylvania, eight other states and the Florida Territory defaulted between 1841 and 1843:

- **Maryland**: defaulted with over $12 million in debt, primarily for canal construction.

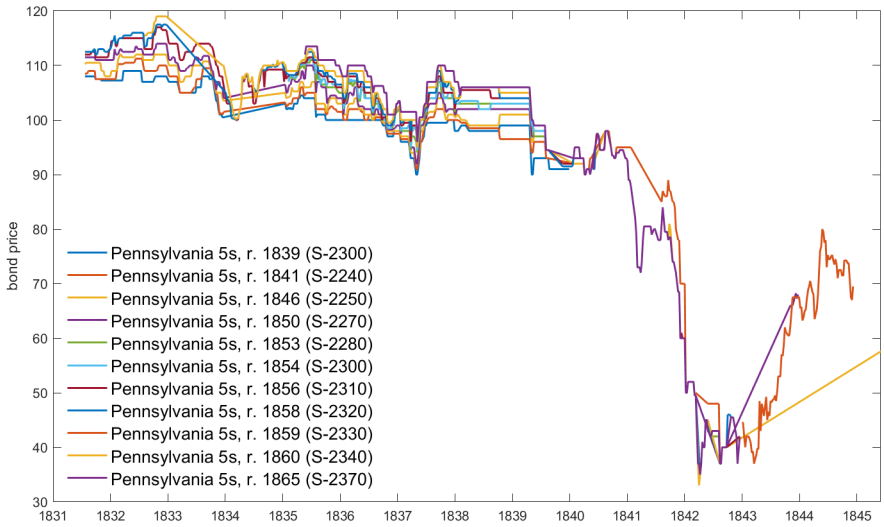

Figure 22: Prices of Various Pennsylvania State Bonds

Source: Sylla et al. (2006).

- **Midwest states** (Indiana, Illinois, Michigan): combined debt of approximately $37 million for railroads and canals.
- **Southern states** (Arkansas, Louisiana, Mississippi): borrowed primarily to capitalize state banks.

Four states ultimately repudiated all or part of their debts, while three underwent substantial renegotiations (Grinath et al. 1997).

5.2 Patterns of Default

Wallis (2005) documents that nine of the ten states with the largest per capita debts defaulted; only New York, Ohio, and Alabama narrowly avoided this fate. The pattern is clear: states that had pursued large-scale infrastructure investment through debt financing, without corresponding tax increases, were most vulnerable when the post-Panic depression reduced toll revenues and land values.

Wallis et al. (2004) highlights a critical distinction between states that defaulted and states with comparable debt levels that avoided default. Maryland and Pennsylvania both had the *fiscal capacity* to service their debts through property taxation; the problem was that both were dilatory in levying adequate taxes. When crisis forced action, Maryland and Pennsylvania enacted property taxes and resumed payments with back interest almost immediately. By contrast, the northwestern states (Indiana, Illinois, Michigan) had already implemented high property taxes and genuinely lacked resources—they were forced to default and partially repudiate.

The New York–Pennsylvania comparison is especially instructive. Both states borrowed heavily for canals; both faced fiscal stress in the early 1840s. The difference lay partly in project quality (Erie Canal revenues substantially exceeded Pennsylvania Main Line revenues) and partly in the pre-existing fund structure that gave New York a balance-sheet cushion that Pennsylvania lacked.

6 Constitutional Responses

State legislatures throughout the country asked "how did we get into this mess?" and "how can we prevent this from happening again?" Although conditions in every state were unique, the constitutional reforms of the 1840s shared a common theme: states got into trouble because they pursued taxless finance, and the solution was to eliminate this option by requiring taxes to be raised when spending is contemplated (Wallis 2005).

6.1 New York Constitution of 1846

New York held a constitutional convention in 1846 that established what scholars call "procedural debt restrictions" in Article VII:

1. A debt limit of $1 million for ordinary purposes (Section 1).
2. For debt exceeding this limit (Section 3), the legislature must:
    - Contract the debt for a specific, stated purpose;
    - Enact a tax sufficient to pay the interest and retire the principal within 18 years;
    - Submit the proposed tax to voters for approval in a statewide referendum.
3. Exceptions only for extraordinary emergencies: invasion, insurrection, or war.

The Harvard Business School case study (Moss and Grodzins 2016) describes the proposal by Michael Hoffman as placing "tight restrictions on state debt, which had increased sharply over the previous eight years." By requiring voter approval of taxes *before* projects were undertaken, the constitution fundamentally changed the political calculus of infrastructure investment.

The procedural restrictions did not limit the *amount* of debt a state could issue; rather, by addressing the timing issue, they created what modern public finance literature would call a hard budget constraint—making it impossible for project promoters to promise valuable infrastructure while deferring the tax burden to the future (Grinath et al. 1997).

The Constitution of 1846 also mandated the creation of a sinking fund for the gradual redemption of the General Fund debt; the State retired the last of that debt in 1878.

6.2 Pennsylvania Constitutional Amendments of 1857

Pennsylvania's constitutional response was more delayed. The state's 1838 constitution did not include comprehensive debt restrictions; major fiscal reforms came as amendments in 1857, fifteen years after the default.

Article XI of the amended Pennsylvania Constitution established:

- **Section 1**: Limited state debt to $750,000 for "casual deficits or failures in revenues."
- **Section 3**: "Except the debts above specified . . . no debt whatever shall be created by, or on behalf of the State."
- **Section 4**: Created a mandatory sinking fund of at least $250,000 annually to retire existing debt.
- **Section 5**: "The credit of the Commonwealth shall not in any manner, or event, be pledged, or loaned, to any individual, company, corporation, or association."
- **Section 6**: Prohibited assuming debts of local governments or corporations.
- **Section 7**: Prohibited counties and cities from becoming stockholders in corporations or lending their credit.

These provisions effectively prohibited Pennsylvania from the practices that had led to default: borrowing without dedicated revenue sources, investing in or lending to private corporations, and assuming the debts of other entities.

6.3 Common Elements Across the Reforms

The constitutional reforms in New York, Pennsylvania, and other states shared several features that together eliminated taxless finance:

Strict Debt Limits

Both states imposed absolute limits on state borrowing for ordinary purposes. New York's $1 million limit and Pennsylvania's $750,000 limit were intentionally restrictive, requiring any significant borrowing to meet extraordinary criteria or undergo special approval.

Prohibition on Lending State Credit

Both states prohibited the state from pledging its credit to, or becoming a stockholder in, private corporations—closing the channel through which they had accumulated contingent and direct debt obligations.

Dedicated Revenue Requirements

Authorized debt had to be accompanied by specific, dedicated tax revenues sufficient to service and retire it, forcing legislators and voters to confront the true cost of projects at the time of authorization.

Voter Approval

New York's requirement for a statewide referendum on debt-servicing taxes was particularly innovative, ensuring that borrowing decisions were made with full awareness of fiscal consequences.

7 The "Taxless Finance" Problem

Wallis (2005) argues that the fundamental problem was not the infrastructure investments themselves—many projects, particularly New York's Erie Canal, were economically sound and generated substantial social returns. The problem was the political economy of deferred taxation.

When project promoters could promise infrastructure while claiming it would "pay for itself" through tolls or increased land values, voters and legislators systematically underestimated risks. Without the requirement to raise current taxes, several problems emerged:

1. **Optimism bias**: Toll revenue projections were systematically optimistic. Pennsylvania's canal revenues never came close to projections, and the state ultimately received only $11 million in sale proceeds for a system that cost $84–87 million to build, finance, and operate.

2. **Risk underestimation**: The probability of adverse economic shocks—such as the Panic of 1837—was not adequately factored into borrowing decisions.

3. **Political agency problems**: Legislators could claim credit for visible infrastructure projects while deferring the fiscal costs to future administrations. Both New York and Pennsylvania exhibit this pattern: the debts accumulated in the 1820s and 1830s came due in the administration of officials who had not made the original borrowing decisions.

The lesson the states drew in the 1840s was that "taxes must be raised when spending is contemplated." If current taxes are not raised, taxpayers and politicians may not adequately factor in the risks of higher taxes in the future (Grinath et al. 1997).

The contrast between the two states' experiences with unit-of-account management reinforces this interpretation. New York's Comptrollers consistently prioritized maintaining the state's credit standing—paying specie rather than depreciated paper, even at short-run fiscal cost—while Pennsylvania's management was more opportunistic. Fiscal discipline and creditworthiness are complementary.

8 Long-Run Impact

The constitutional restrictions adopted in the 1840s and 1850s proved remarkably durable. No state has experienced a full default since Arkansas in 1933, despite numerous economic downturns over the intervening nine decades.

The reforms established several enduring principles in American state public finance:

- **Balanced budget requirements**: Most state constitutions now require balanced operating budgets.
- **Debt limitations**: States maintain various forms of constitutional or statutory debt limits.
- **Voter approval for borrowing**: Many states require voter approval for general obligation bonds.
- **Separation of state and private enterprise**: States generally cannot invest in or lend to private companies.

These restrictions have been criticized at times for limiting states' flexibility in responding to economic conditions or funding infrastructure. They represent a deliberate choice, made in the wake of fiscal crisis, to constrain the ability of political actors to defer fiscal burdens to the future.

9 Conclusion

The state debt crisis of the 1840s and the constitutional responses it provoked represent a critical episode in the development of American fiscal federalism. New York and Pennsylvania—the "good" and "bad" case studies—illustrate how similar fiscal strategies (canal-financed through debt, without commensurate taxation) led to different outcomes depending on project quality, balance-sheet cushions, and willingness to tax.

Pennsylvania's default was not inevitable: the Commonwealth had the fiscal capacity to service its debt through property taxation. The problem, as Wallis et al. (2004) emphasizes, was the political economy of deferred taxation—the unwillingness to impose current taxes in the face of projected (but ultimately unrealized) toll revenues.

The constitutional reforms adopted by New York, Pennsylvania, and other states addressed this problem by eliminating taxless finance: requiring that taxes be approved at the time debt is incurred, and prohibiting state investment in or guarantees of private enterprise. These institutions have proven remarkably stable and effective, shaping state fiscal policy nearly two centuries later.

10 Data Sources

New York

- Revenue and expenditure data for the General Account: Appendices I and II of Sowers (1914) and the *Annual Report of the Comptroller* for the State of New York.
- Debt outstanding, fund balances, and Treasury balances: *Annual Report of the Comptroller*.
- Canal Account revenues, expenditures, debt outstanding, and fund balances: Table 82, p. 88 of the 1860 *Annual Report of the Commissioners of the Canal Fund to the Legislature of the State of New York*.
- Bond prices: Sylla et al. (2006).
- Bank Safety Fund details: Chaddock (1910) and Bodenhorn (1996); general fund history: Roberts (1897).

Pennsylvania

- Revenue, expenditure, state balance, and debt data: various issues of the *Report on the Finances of the Commonwealth of Pennsylvania Made to the Legislature by the Auditor General* and the *Report of the State Treasurer upon the Finances of Pennsylvania*.
- Canal and railroad cost and revenue estimates: Pennsylvania Office of the Auditor General and Treasury (1854).
- Debt ownership: Pennsylvania Office of the Auditor General (1842).
- Bond prices: Sylla et al. (2006).
- Default chronology: English (1996).

In both cases our fiscal records are consistent with the budget constraints presented in Section 2.

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